

Customer Churn Insight Report

Customer-facing summary based on model scores and explanation outputs

1) Executive Summary

This report highlights the strongest churn drivers identified by the model and the highest-risk customers in the test set. The goal is to help retention teams prioritize outreach based on predicted churn risk and the most influential customer attributes.

Total explained customers: **1,409** | Average predicted churn probability: **0.394**

2) Main Churn Drivers

Feature	Avg Impact	Avg Direction
tenure	0.0894	raises risk
InternetService_Fiber optic	0.0869	raises risk
MonthlyCharges	0.0559	raises risk
TotalCharges	0.0528	raises risk
Contract_Two year	0.0369	lowers risk
InternetService_No	0.0266	lowers risk
PaymentMethod_Electronic check	0.0227	raises risk
Contract_One year	0.0200	lowers risk

3) Highest-Risk Customers

customerID	pred_proba	pred_label	top_features	top_shap_values
1069-XAIEM	0.960	1	tenure;InternetService_Fiber optic;MonthlyCharges	0.2420;0.1897;0.0833
9248-OJYKK	0.958	1	InternetService_Fiber optic;tenure;TotalCharges	0.2413;0.2334;0.1060
5178-LMXOP	0.955	1	tenure;InternetService_Fiber optic;MonthlyCharges	0.2191;0.2165;0.0894
9804-ICWBG	0.944	1	InternetService_Fiber optic;tenure;TotalCharges	0.2693;0.2540;0.1719
5542-TBBWB	0.944	1	InternetService_Fiber optic;tenure;TotalCharges	0.2693;0.2540;0.1719
8375-DKEBR	0.944	1	tenure;InternetService_Fiber optic;TotalCharges	0.2650;0.2005;0.1869
1820-TQVEV	0.943	1	InternetService_Fiber optic;tenure;TotalCharges	0.2690;0.2500;0.1860
0295-PPHDO	0.943	1	InternetService_Fiber optic;tenure;MonthlyCharges	0.2169;0.2076;0.0844
2609-IAICY	0.940	1	InternetService_Fiber optic;tenure;TotalCharges	0.2696;0.2518;0.1148
2514-GINMM	0.940	1	InternetService_Fiber optic;tenure;TotalCharges	0.2914;0.2386;0.1245

4) Retention Guidance

Start with customers showing the highest predicted churn probability. Focus outreach on the top drivers that push risk upward, such as month-to-month contracts, fiber-optic service, and electronic check payments. For lower-risk customers, use automated nurture campaigns instead of direct discounting.